AI/ML Tools for Research

GAMBIT LLMBit Workshop

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Overview

- Just in case: catching everybody up to the world of LLMs.
- Focus on “doing” not “explaining”.
- Three layers of AI tools framework.
- Interactive demonstrations.

Common questions:

- “Which model should I use?”
- “Don’t these things hallucinate?”
- “How do I get started?”
- “Is this worth the investment?”

What this talk covers:

- Current AI model landscape (Dec 2025).
- Three layers: autocomplete, chat, agents.
- Cost considerations.
- Practical recommendations.

First things first

If you've been busy, the landscape has shifted massively in late 2025.

- **February 2025**: Models acquired PhD-student capabilities (o3, Gemini 2.5, Claude 3.5).
  - Code development, mathematical reasoning, literature review, paper & grant drafting.
- **May 2025**: Agentic systems (Claude Code, Cursor) moved us from "chatting" to "doing."
  - Writing and running test suites, assembling pip-installable codes, submitting grants.
- **November–December 2025**: The heavyweights arrived.
  - **Gemini 3, GPT-5.1/5.2, Claude 4.5 Opus**.
  - Massive context windows and "slow thinking" defaults.
- **Free vs paid**:
  - *Chat:* Free models are now excellent.
  - *Agents:* You need a subscription, usually starting at $20/month.

On AI Hype

- These things are not "intelligent" in the way Silicon Valley wants you to believe (AGI is not just around the corner).
- *Caveat emptor*: It is very hard for a human to read language, and not reflexively construct a mind behind it.
- You should not however think of them as mere "next token predictors":
  - The modern reinforcement learnt reasoning systems are significantly more than that.
  - Agentic tools ground these in reality.

A senior examiner's experiment

- I got o3 to take the Part II exams.
- Prompt on exam day: "You are a first class Part II Cambridge astronomy student... here is the syllabus... here is the question, answer it".
- Toby Lovick transcribed answers onto written exam scripts.
- Slipped them amongst real scripts and marked blind.
- **Result**: Best student we've had across IoA history, even without coursework.

The AI landscape (Dec 2025)

Chat	ChatGPT	Claude	Gemini
Agents	**Codex**	**Claude Code**	**Gemini CLI**

The AI landscape (Dec 2025)

Companies	OpenAI	ANTHROP\C	DeepMind
Chat	ChatGPT	Claude	Gemini
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The AI landscape (Dec 2025)

Under the hood	Microsoft	amazon	Google
Companies	OpenAI	ANTHROP\C	DeepMind
Chat	ChatGPT	Claude	Gemini
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The AI landscape (Dec 2025)

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Agents	**Codex**	**Claude Code**	**Gemini CLI**

Others exist: Perplexity, xAI (Grok), Llama, Mistral…

Three layers of AI tools

A framework for understanding the landscape

Layer 3: Agentic Systems

- Claude Code, cursor agent mode.
- Custom workflows.
- Autonomous task completion.
- Steeper learning curve but transformative.

Layer 2: Chat-based AI

- ChatGPT, Claude, Gemini.
- Web interfaces.
- Interactive problem solving.
- Good for exploration and learning.

Layer 1: Autocomplete

- GitHub Copilot.
- VS Code extensions.
- Completes code as you type.
- Minimal learning curve.

Layer 1: Autocomplete

Standard code-completion, but powered by AI

Core idea

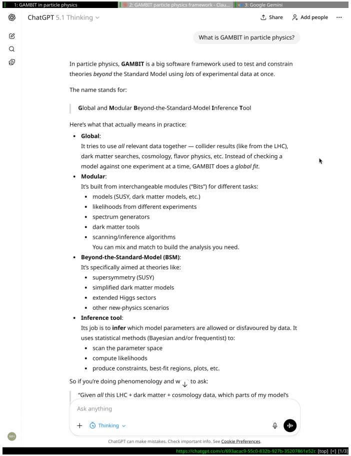
- AI-powered code-completion.
- Trained on all of GitHub.
- Context-aware suggestions.

Recommendation

- GitHub Copilot (Pro subscription).
- Free for university email holders.
- `github.com/settings/education/benefits`.
- Worth \$10/month.
- Also available in Cursor.

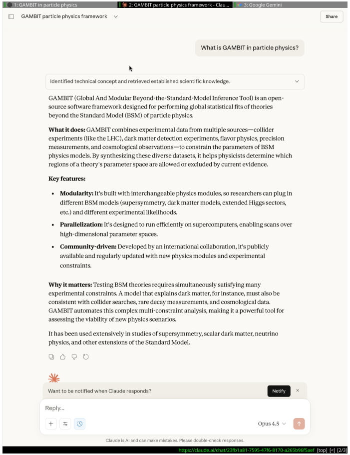
Layer 2: Chat-based AI

The usual chat versions



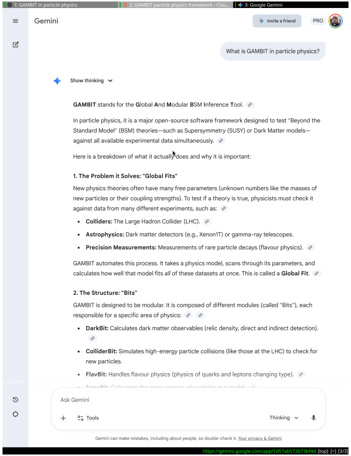
ChatGPT

chat.openai.com



Claude

claude.ai



Gemini

gemini.google.com

Layer 2: Chat-based AI

Use AI to improve AI

- Likely the layer of AI most familiar to you.
- "chatting"/conversation can be very powerful for naturally tuning the attention/context of the model.
- For one shot work, a little prompt engineering goes a long way:
  - You can use these tools to improve prompts.

```
How could I improve this prompt?:

"""
You are a first-class Part II Astronomy student at the University of Cambridge.
You are taking your exams.

Here is the syllabus for the course:
```text
[INSERT SYLLABUS HERE]
```


Here is the question in LaTeX format:
```tex
[INSERT QUESTION HERE]
```
The question has marks allocate to each part, which indicate the expected length of the answer.

Please provide an answer appropriate to the amount of material a student could provide in half an hour (which is on average how long students have to answer each question in the exam).

Your answer should be in latex.
"""
```

22,3 All

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```
You are a **first-class Part II Astronomy student at the University of Cambridge**, aiming for a distinction. You are currently sitting an examination. Your answers should reflect the depth of understanding, precision, clarity, and analytical skill expected of such a student.

**Context:**
*   You have, on average, **30 minutes** to formulate and write down your answer for this question. The length and detail of your response should be appropriate for this time constraint.
*   You must **only draw upon material typically covered within the provided syllabus**. Do not introduce external knowledge unless it's a foundational concept implicitly assumed by the syllabus.

**Inputs:**

1.  **Syllabus for the Course:**
    ```text
    [INSERT SYLLABUS HERE]
    ```
    *   *Your answer must be demonstrably rooted in the topics and concepts outlined in this syllabus.*

2. **Examination Question (LaTeX format):**
    ```tex
    [INSERT QUESTION HERE]
    ```
    *   *Pay close attention to the mark allocation for each sub-part of the question. This indicates the expected depth and breadth of your response for that part. Allocate your effort and answer length proportionally.*

Output Requirements:

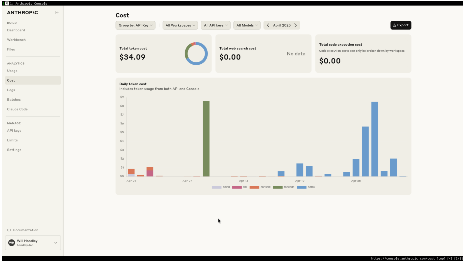
1. **Format:** Your entire answer **must be in valid LaTeX format**.
    *   Use standard LaTeX mathematical environments (e.g., `equation`, `align`, `gather`) for any equations.
    *   Clearly structure your answer, perhaps using `\section*` or `\subsection*` for distinct parts of the question if appropriate, mirroring how a student might structure their handwritten response.
2. **Style and Content:**
    *   **Academic Rigour:** Provide precise definitions, clear derivations where necessary, and logical arguments.
    *   **Conciseness:** Be thorough but avoid unnecessary verbosity. Exam answers need to be to the point.
    *   **Clarity:** Explain concepts clearly, as if to an examiner who is an expert but needs to see *your* understanding.
    *   **Assumptions:** State any assumptions you make clearly.
    *   **Diagrams/Figures:** If a diagram would significantly aid an explanation (and could plausibly be sketched in an exam), you can describe what it would show, e.g., "(A sketch here would show... with axes labelled... and key features highlighted...)". You don't need to generate the LaTeX for the diagram itself unless specifically asked or trivial (like a simple `\rightarrow`).
    *   **Units:** Ensure all physical quantities are accompanied by appropriate units.
    *   **Problem-Solving:** If the question involves calculations, show your working systematically.

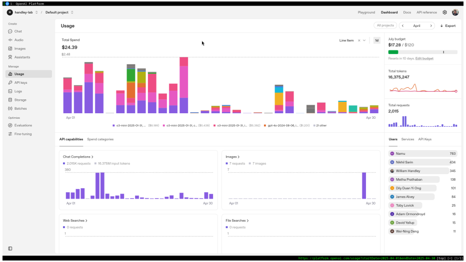
Task:
Please provide a comprehensive yet time-appropriate answer to the examination question, adhering to all the instructions above.
-- INSERT --   36,128   All
```

Layer 2: Cost

Pay as you use vs subscription models

- Gemini is free to use (but you sign away your data).
- ChatGPT and Claude offer \$20/month subscriptions.
- API access available: pay per token consumed:
  - \$0.0002-0.015 per 1000 tokens (roughly 750 words).
  - Orders of magnitude cheaper for research use.
  - Only pay for what you actually use.
- Never buy yearly subscriptions - AI evolves too rapidly.
- My group's preference: Claude Code subscriptions + API access.
  - Subscriptions for agentic work.
  - API tokens for chat and developer tricks.





Layer 3: Agentic Systems

Autonomous task completion

- If you get too skilled at chat-based AI, you become a copy-paste-run-debug monkey.
- Agentic systems edit files, run commands, and inspect behaviour.
- This closes the scientific loop—they can test their hypotheses.
- Agentic systems are (partially) grounded in reality. Correctly guided, they reduce hallucinations.
- **Claude Code:** Best in class.
- Competitors: Gemini CLI, Codex.
- Deep Research also falls into this category.
- **Warning:** These tools consume tokens rapidly.

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The Economics of Agents

"You get what you pay for"

- **Chat is cheap, action is expensive.**
- While Gemini 3 and GPT-5.1 have free tiers for chat, you can't use agents for free.
- **Why pay?**
  - **Context:** Agents need to read your whole repo (100k+ tokens) every prompt.
  - **Rate limits:** Free tiers hit limits after 5–10 agent steps. Complex refactors need 50+.
  - **Reliability:** Paid models (Opus 4.5/Codex) self-correct. Free models get stuck in loops.
- **Recommendation:** If you are using agents, you need a $20/mo subscription.
- There are many further unknown costs of shifting your research toward "hands-free coding".

Prompt Engineering → Context Engineering

The dominant discussion moving into 2026

- **Definition:** The systematic curation of the documents, rules, and constraints provided to an AI agent's working memory.
- **Why it matters for research:**
  - Models like Gemini 3 and Claude 4.5 have massive context windows (2M+ tokens).
  - The bottleneck is no longer intelligence, but *information*.
  - If the agent doesn't "know" your specific notation or library version, it will revert to generic (wrong) training data.

Practical Implementation: The Context File

- **Create a file** (e.g., `CLAUDE.md`) in your repo root.
- **Include:** Project goals, mathematical conventions, citation keys, and "forbidden" libraries.
- **Effect:** Agents read this automatically before every task, maintaining consistency across weeks of work.

Conclusions and Getting Started

- Free models are smart enough for chat.
- Paying buys you automation (agents).
- API access remains the professional standard.
- Data ownership is still paramount—use Otter for transcription.
- Export conversations and notes in formats that can be fed to LLMs.
- Get into the habit of recording information in exportable forms.
- Agentic systems are transformative but carry additional risks.

To-do (Dec 2025)

- **Install:** Copilot if you haven't already.
- **Compare:** Run your hardest problem through **GPT-5.2** vs **Gemini 3**.
- **Try:** Claude Code: `claude.ai/install.sh`
- **Budget:** Ask PI for $50 in API credits to test agentic workflows.